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Soybean

Growth and Management

QUICK GUIDE

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Growth, development and seed yield of soybeans are a result of a variety's genetic potential interacting with environmental and production management practices.

It is important for successful soybean production to understand plant growth staging and timing when making production decisions. Minimizing environmental stress, particularly during critical stages of soybean development, will optimize seed yield. Management practices that may influence crop growth include variety selection, field selection, seedbed preparation, plant establishment (planting date, rate, depth and row width), pest management (diseases, insects and weeds) and plant nutrition.

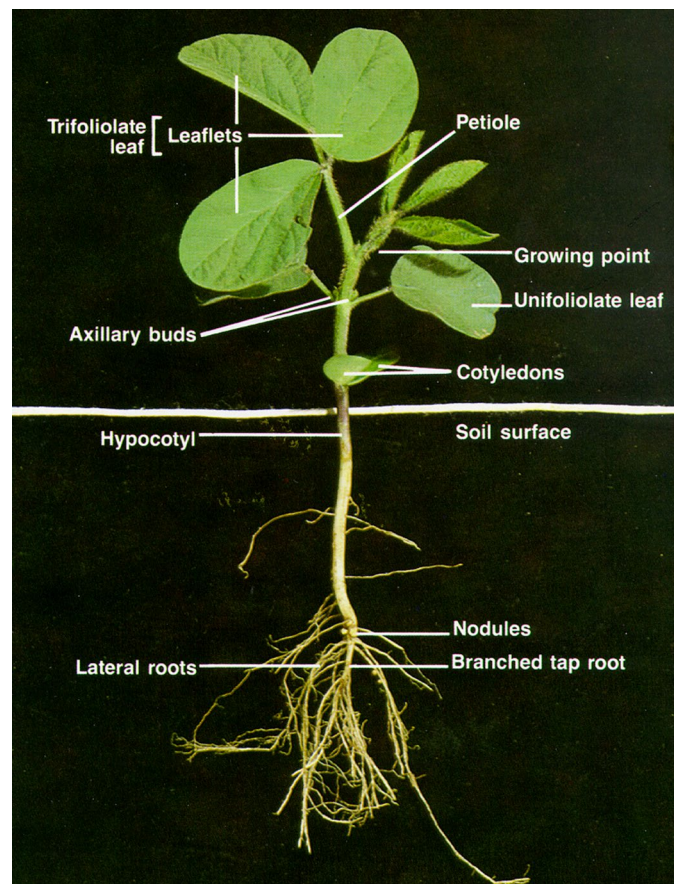


Figure 1.
Soybean plant.

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Soybeans are classified as either indeterminate or determinate in growth. Many southern varieties are determinate in growth and cease vegetative growth when the main stem terminates in a cluster of mature pods. Most northern (including North Dakota) varieties are indeterminate in growth habit. Indeterminate varieties develop leaves and flowers simultaneously throughout a portion of their reproductive period, normally ending during the early seed development growth stage. With soybean development being driven by photoperiod (interval in a 24-hour period during which a plant is exposed to light) and temperature, northern varieties have vegetative growth limited by the season length.

Shortening day length (increasing night length) and warm temperatures control soybean flowering. Soybeans in the northern U.S. have long minimum day-length requirements for the onset of flowering (often greater than 14 hours of daylight). However, even within a variety, variations in time of flowering may occur from year to year, with the same day length closely associated with temperature conditions.

Soybean maturity groups are based on adaptation within certain latitudes. These maturity belts run east to west in the U.S., with only about 100 to 150 miles from the north to the south of each belt. Maturity groups range from 00 in the extreme northern U.S. to VIII in the southern Gulf Coast states and most of Florida.

Planting a specific variety farther north than its adapted maturity range will extend the period of vegetative growth and delay flowering and maturity due to the extended summer day length and cooler temperatures. Likewise, planting a variety farther south than its adapted range will shorten the vegetative growth period, cause earlier flowering and result in an earlier maturity due to shorter summer day length and warmer temperatures.

Vegetative Stages

VE (emergence)
VC (cotyledon stage)
V1 (first trifoliolate)
V2 (second trifoliolate)
V3 (third trifoliolate)
V(n) (nth trifoliolate)

Reproductive Stages

R1 (beginning bloom, first flower)
R2 (full bloom, flower in top two nodes)
R3 (beginning pod, 3/16-inch pod in top four nodes)
R4 (full pod, 3/4-inch pod in top four nodes)
R5 (1/8-inch seed in top four nodes)
R6 (full-size seed in top four nodes)
R7 (beginning maturity, one mature pod)
R8 (full maturity, 95% of pods on the plant are mature)

The vegetative stages are defined by the number of nodes with fully expanded leaves. All V1 to Vn stages have true leaves (leaflet edges no longer touching) that are trifoliolate and produced singularly on different nodes, with these leaves alternating on the stem.

The reproductive stages are divided into four parts: R1 and R2 describe flowering, R3 and R4 describe pod development, R5 and R6 describe seed development and R7 and R8 describe plant and seed maturation.

Vegetative

Vegetative Growth Stages

1. Emergence (VE)

Soybean seed begins germination when the water absorbed is equal to about 50% of the seed's weight. The radical, or primary root, is first to emerge from the seed. Shortly afterward, the hypocotyl (stem) emerges and begins growing toward the soil surface, pulling the cotyledons (seed leaves) with it. This hook-shaped hypocotyl straightens out once emerged and as the cotyledons unfold. As the hypocotyl arch is exposed to light and straightens to pull the cotyledons out of the ground, epicotyl growth begins with expansion and unfolding of unifoliolate leaves. Auxiliary buds are located where the cotyledons or leaves are attached to the stem. Plant emergence normally takes 10 to 18 days, depending on soil temperature and moisture, variety and planting depth. During this time, lateral roots also begin to grow from the primary root. Root hairs also begin to emerge and provide the key nutrient and water-absorbing functions of the plant.

The primary root will continue growing and branching so that lateral roots can reach the center of a 30-inch row within five to six weeks. Eventually, **soybean roots normally reach a depth of 2 to 3 feet, with most of the roots in the upper 6 to 12 inches of soil.**

Soybeans should be **planted 1 to 1½ inch deep but not deeper than 2 inches.**

Low amounts of starter fertilizer (phosphorus, if at less than medium soil levels) in a band within 2 inches of the seed likely will promote early plant growth and seed yield. Do not place fertilizer directly in the furrow because salt (and ammonium, if fertilizer contains nitrogen) injury from the fertilizer is likely to result. Soybeans are very salt sensitive (about twice as sensitive as corn).

If fields have not been in soybeans during the previous three years or more, consider inoculating seed with *Bradyrhizobium japonicum* bacteria to form nodules on the soybean roots that later will provide much of the plant's nitrogen supply.

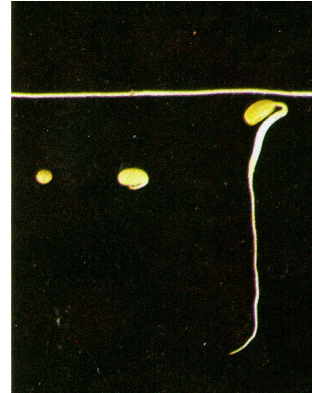


Figure 2. Germination.



Figure 3. VE soybean emergence.

Vegetative Growth Stages



Figure 4. VC soybean.



Figure 5. V2 soybean.

2. Cotyledon Stage (VC)

The VC stage is reached when the unifoliolate leaves are fully expanded. During the VC stage, the cotyledons supply the nutrient needs of the young plant (for about seven to 10 days). If one cotyledon is lost during this time, it has little effect on the plant's growth rate. However, losing both cotyledons at or soon after VE can reduce yields by up to 10%. Later, after V1, photosynthesis in the developing leaves allows the plant to sustain itself.

3. First trifoliolate (V1)

This stage is achieved when the first trifoliolate is fully emerged and opened. The V stages after VC are defined and numbered by the upper, fully developed leaf node on the main stem above the unifoliolate leaves (the V1 to Vn stages are numbered by fully developed trifoliolates). New V stages will begin appearing about every three to five days through V5 and then every two to three days from V5 to shortly after R5, when the node number usually reaches a maximum. Trifoliolate leaves on branches are not counted when determining V stages; only the trifoliolates off the main stem are used in the count.

4. Second node (V2)

Plants have two nodes above the unifoliolate node, with trifoliolate leaves having unfolded leaflets. **Active nitrogen fixation is just beginning to occur.** Most of these root nodules are within 10 inches of the soil surface, with millions of bacteria in each nodule.

Nodules that are pink or red inside are active in nitrogen fixation. White, brown or green nodules are not efficiently fixing nitrogen and are probably parasitic on the plant. The plant will use bacteria-produced nitrogen and the applied or intrinsic soil nitrogen. Too much soil nitrogen will cause the plant to use this N supply first, delaying nodulation and/or reducing nodule size and number.

Lateral roots are developing rapidly in the top 6 inches of soil.

Vegetative

Vegetative Growth Stages

5. Third to Fifth nodes (V3-V5)

Soybean plants have three nodes above the unifoliolate node at V3, and five nodes of fully expanded leaflets at V5. The number of branches seen on the plant may increase at this point in wider row spacings and under lower plant densities, depending upon the variety grown.

Normally at V5, the plant's axillary buds in the top stem will develop into flower clusters (racemes). V5 is about one week or less from R1, or first flower. At V5, the total number of nodes that the plant can produce is established.

Although the stem apex (main growing point) is dominant, damage to this growing point will allow axillary buds lower on the plant to branch and grow profusely. Thus, soybeans are capable of producing new branches and leaves after hail destroys almost all of the above-ground foliage as long as at least one axillary bud remains intact. However, **if the plant is broken off below the cotyledonary node, the plant is killed** because no axillary buds are below this node.

6. Sixth node (V6)

Soybeans at this stage have six nodes containing trifoliolate leaves with fully unfolded and expanded leaflets. The unifoliolate and cotyledons may have senesced from the plant. The plant can readily recover from damage; a 50% leaf loss at this stage will reduce yield less than 5%.



Figure 6. V5 soybean.



Figure 7. V6 soybean.

Reproductive

Reproductive Stages



Figure 8. R1 soybean.

1. Beginning Bloom (R1)

At least one flower is on the plant at any node on the main stem. Soybean **flowering always initiates on the third to sixth node** on the main stem, depending on vegetative stage when flowering begins. This flower initiation will progress up and down the plant. Branches eventually also flower. Within each raceme, the flowering will occur from the base to the tip, so basal pods are always more mature.

Dominance of the primary racemes is seen over secondary racemes on the plant; however, secondary racemes can develop just to the side of primary racemes on the same axil. Vertical roots are rapidly growing and will continue until R4 to R5, as are secondary roots and root hairs nearer the soil surface.



Figure 9. R2 soybean.

2. Full Bloom (R2)

An open flower is seen at one of the two top nodes of the main stem. At least one of these two upper nodes shows a fully developed leaf. At this stage, **the soybean has accumulated about 25% of its total dry weight and nutrients** and has obtained about **50% of its mature height**. About 50% of the total mature node number has been established. **Very rapid nitrogen, phosphorus, potassium and dry-matter accumulation is occurring and will continue through R6.**

The appearance of new flowers on the plant begins to slow between R2.5 and R3 and **will be complete by R5**. Major lateral roots have turned downward in the soil and nitrogen fixation by root nodules is increasing rapidly. Defoliation of the plant of 50% at this stage will reduce yield less than 10%.



Figure 10. R3 soybean.

3. Beginning Pod (R3)

A pod on the upper four nodes is $\frac{3}{16}$ inch long. Temperature or drought stress **at this time can affect yield through total pod number, bean number per pod or seed size**. Partial compensation with only temporary stress can occur in soybeans, but as the plant matures from R1 to R5.5, this ability to compensate will decrease.

Very favorable conditions will result in greater pod numbers per plant at this time. Any stress at this stage will increase pod and flower abortion, impacting yield greatly. However, the long flowering period of soybeans is one reason the plants can compensate so well.

4. Full Pod (R4)

This stage shows rapid pod growth and the beginning of seed development at the beginning of the full-pod stage. The dry weight of pods increases greatly from R4 to R5. The plant has a pod at this stage that is at least 3/4-inch long on at least one of the four upper nodes of the main stem. The last flowering will occur at the main stem tip (through R5).

This stage is the most crucial period for seed yield. Any stress from R4 to R6 causes more yield reduction than at any other time. Late pod formation at R4.5 to early seed fill at R5.5 is most critical. Yield reduction at this time is mainly from fewer pods. This is a critical period to have implemented all management practices for preserving yield.

5. Beginning Seed (R5)

Seed development during this stage requires much water and nutrients from the plant. Redistribution of nutrients in the plant occurs with the soybean providing about half of the needed N, P and K from the plant's vegetative parts and about half from N fixation and nutrient uptake by the roots from the soil.

Leaf loss of 100% at this stage will reduce yield by 50-80%; the plant is less able to compensate for stress and vegetative damage. Stress can lower yields by reducing pod numbers and the number of beans per pod, and to a lesser extent, by reducing seed size.

This stage has a pod on one of the four upper nodes of the main stem with a seed at least 1/8 inch long. **About halfway through this stage, the plant attains its maximum height, node number and leaf area. Nitrogen fixation peaks and begins to drop, and the seeds continue a steady period of dry weight accumulation.** Toward the end of this stage, the nutrient accumulation in the leaves peaks and then begins the process of redistributing to the seed.

Seed nutrient accumulation will continue until shortly after R6.5, with about 80% of total seed dry weight being accomplished.

6. Full Seed (R6)

This stage is also known as the “green bean” stage or beginning full-seed stage, and total pod weight will peak during this stage. Within this stage, three to six trifoliolate leaves may fall from the lowest nodes on the plant prior to leaf yellowing. This stage initiates with a pod containing a green seed that fills the pod cavity on at least one of the four top nodes of the main stem. **The growth rate of the beans is rapid but will slow by R6.5 and peak at R7.**

Root growth is complete at about R6.5.



Figure 11. R4 soybean.



Figure 12. R5 soybean.



Figure 13. R6 soybean.

Reproductive Stages



Figure 14. R7 soybean.



Figure 15. R8 soybean.

7. Beginning Maturity (R7)

This stage begins with one normal pod on the main stem, which obtains the mature color (brown or tan). Dry matter begins to peak in individual seeds. This is visible when all the green is lost from the seeds and pods (they appear yellow). **Leaf yellowing and loss begins at this stage.**

Seeds contain about 35% moisture at physiological maturity. Stress at this stage or later has almost no effect on yield unless pods are dropped to the ground or seeds are shattered from the pods. Also, any lodged plants may reduce actual yield (due to reduced light interception) and harvested yield (harvest losses). **At this stage, the crop is safe from a killing frost.**

8. Full Maturity (R8)

On the soybean plant, 95% of the pods have reached their mature color. Generally, 5 to 10 days of good drying weather after this stage will be required to have the soybeans at less than 15% moisture, or harvest moisture. Soybeans will lose moisture rapidly with warm and dry weather at this point and should be timely harvested to prevent losses.

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Soybean is a very important legume crop in India. It contributes 45% of the total oilseeds and about 25% of the total edible oil produced in the country and the seed contains 40% protein and 20 % oil being one of the most economical sources of good quality protein. It also contains many minerals and useful nutraceuticals like iso-flavones, which have immense health benefits. Therefore, the crop has a potential to provide nutritional security and eradicate rampant protein malnutrition in the country. Soybean being a leguminous crop fixes atmospheric nitrogen, reduces consumption of synthetic nitrogenous fertilizer in the subsequent crop and maintains soil fertility.

Agronomic practices

Soil Requirement

Soybean can be grown in variety of soil types. However, a well-drained, sandy loam to clayey soils with medium water holding capacity, rich in organic carbon and leveled fields with near neutral pH is ideal for harnessing maximum soybean yield. Soil with excessive salts/ sodium, acidic and poorly drained soil are not suitable for soybean. It is noteworthy that the crop is predominantly grown under black cotton areas at present.

Tillage

Deep summer ploughing using reversible Mould board Plough, facilitates exposing the hibernating insects to extreme heat and predatory birds as well as movement of nutrients and infiltration of soil water. Therefore, one deep ploughing once in 3-4 years, otherwise one normal ploughing in summer followed by 2 criss-cross harrowing or cultivation for breaking of soil clods will make ideal seed bed for a good crop of soybean cultivation is recommended. Also, sub-soiling operation once in 4-5 years at an interval of 10 meter, break the compactness of the sub-soil and also facilitate infiltration of rainwater which is useful for un-interrupted crop growth even during drought period also.

Use of organic manure

In order to have sustainable soybean yield, it is very important to maintain soil health. Hence, farmers are advised to incorporate organic manure (well decomposed FYM @ 5-10 t/ha or Compost @ 5 ton/ha or Poultry Manure @ 2.5 t/ha) at the time of land preparation. If the quantity of organic manure is limited, they are advised to apply the same on rotation basis in their fields every year. In case of saline soil (pH >7.5), it is also advised to apply Gypsum @ 150-200 kg/ha along with FYM/Poultry manure before the last harrowing in case the nutritional requirement is to be planned without sulphur sources for maintenance of soil pH. Similarly, in acidic soil (pH <6.5) it is advised to use lime @ 600 kg/ha.

Harrowing

Soon after arrival of monsoon, the land may be prepared by harrowing followed by planking to level the field.

Sowing techniques

With increased mechanization and availability of tractor drawn seed drills, farmers have shifted from bullock drawn sowing equipment like dufan/tifan to use of multi-crop seed drills having adjustable row-to-row distance and seed rate as per the requirement of the crops. Normally these seed drills have provision of flat sowing 5-9 rows of soybean crop with adjustable row to row distance of 14-18 inches. Looking to the prevailing climatic aberrations and erratic rainfall, following methods can be used to mitigate the adverse climate.

- Broad Bed Furrow planting: The BBF seed drills have a provision of opening the irrigation channels after an interval of 4-5 rows. The furrow mechanism is fitted on both the ends of BBF seed drills.
- Ridges & Furrows planting (FIRB (furrow irrigated raised bed) : Seed drill developed by the institute can be used for sowing each row/paired rows on ridges.

Selection of varieties

In order to avoid risk of yield reduction due to aberrant climatic situation and biotic factors, farmers are advised to grow 3-4 soybean varieties with varying maturity periods (Varietal Cafeteria Approach) in their fields. Different varieties possess resistance/tolerance to particular insect-pest and diseases. Long duration soybean varieties are able to give more yields subject to application of irrigation in event of early cessation of monsoon.

Seed treatment and inoculation

Seed treatment is very important operation in soybean considering number considering number of fungal, bacterial and viral diseases which causes considerable reduction in plant population and thereby yield. Hence, farmers are advised to treat soybean seed at the time of sowing using premixed fungicides like Azoxystrobin 2.5% + Thiophanate Methyl 11.25% + Thiamethoxam 25% FS (10 ml/kg seed) or Penflufen + Trifloxystrobin 38 FS (1 ml/kg seed) or Fluxapyroxad 333 g/l FS (1 ml/kg seed) or Carboxin 37.5 + Thiram 37.5 (3g/kg seed) or Carbendazim 25%+ Mencozeb 50% WS (3g/kg seed) or Trichoderma viride @ 8-10 g/kg seed.

In areas where Yellow Mosaic Virus and Stem fly are affecting the soybean crop every year, farmers are recommended to carry out seed treatment with recommended insecticide i.e. Thiamethoxam 30 FS @10 ml/kg of seed or Imidacloprid 48 FS@ 1.25 ml/kg seed.

Once the seed treatment with fungicides and insecticide is done, farmers are advised to inoculate the treated seed with bio-inoculants like

Bradyrhizobium japonicum and Phosphate Solubilizing Micro-organism (PSM) each @ 5 g/kg seed immediately before sowing. If the soybean is grown in nontraditional/new area, they should increase the quantity of bio-inoculants to atleast 10-15 g/kg seed.

Sowing time, spacing and seed rate for soybean

Zone-wise optimum sowing time, seed rate and row spacing for soybean

Zone	Sowing Time	Seed rate (kg/ha)	Spacing (cm)
North Eastern Hill	15th June - 30th June	55	45
North Plain	20th June - 5th July	65	45
Eastern	15th June - 30th June	55	45
Central	20th June-5th July	65	45
Southern	15th June-30th June	65	30

Manures and fertilizers

Fertilizer dose recommended for soybean

Zone	Recommended Fertilizer dose (NPKS kg/ha)	Fertilizer Sources
Central	25:60:40:20	56 kg Urea+375 kg SSP+ 67 kg MoP
Southern	25:80:20:30	56 kg Urea+500 kg SSP+ 34 kg MoP

North Plain	25:75:25:37.5	56 kg Urea+470 kg SSP+ 42 kg MoP
North Eastern Hill	25:100:50:50	56 kg Urea+625 kg SSP+ 84 kg MoP
Eastern	25:100:50:50	56 kg Urea+625 kg SSP+ 84 kg MoP

Intercropping in soybean

Soybean is grown throughout the country as a rainfed crop during the June/July to October. It is also successfully grown as intercrop (4:2 or 2:2 combination) with cotton, maize etc. and also in sugarcane and other horticultural crops.

Weed management

The critical period of weed crop competition in soybean is 45-60 DAS. Therefore, the weeds should be managed during this period. Two hand weeding at 20 & 40 DAS or intercultural operation (Hand hoe/Dora/Kulpa) using bullock drawn/tractor drawn implements during critical period is found beneficial. In situ mulching with weeds at 20 DAS is also beneficial for controlling weeds as well as to conserve soil moisture and add the organic matter in the soil.

Integrated Pest and disease management

- Summer deep ploughing:** Deep ploughing during summer facilitates exposing the hibernating insects to extreme heat and predatory birds.
- Proper sowing time:** Crop sown during June end, escapes damage due to stem fly. Whereas girdle beetle infestation can be minimized by delayed sowing in July
- Optimum/Recommended seed rate, spacing and plant population:** Higher seed rate results in more densely populated soybean crop which attracts more insects. Further, it promotes lodging resulting in yield loss. Hence, the optimum seed rate of (60-80 kg/ha) depending on seed size and germination with recommended spacing (30-45 cm x 5 cm) is recommended.
- Selection of insect resistant/tolerant variety:** Varieties recommended for a particular agro-climatic zone, should only be used. Cultivation of marginally less yielding but insect-resistant or tolerant variety is always remunerative.
- Balanced nutrition:** Use of excessive nitrogenous fertilizers leads to more incidence of defoliators and girdle beetle. Therefore, use optimum quantity of recommended nutrients only. Use of potassium improves crop health and provides resistance against insect-pests.
- Destruction of infested plants:** Bihar hairy caterpillar and tobacco caterpillar are gregarious feeders at initial stage and the infested plants are easily spotted in the field. Removal and destruction of such plants prevents larvae to migrate to and damage other plants. Likewise, girdle beetle infested plants are also easily recognized by dried portion above the girdles. Up to 30-45 days, remove the infested plant parts from below the girdles to destroy eggs and grubs of girdle beetle. Regular scouting and monitoring helps.
- Light trap:** Adult moths of most of the defoliating larvae are nocturnal, and are attracted towards light source. These should be collected with the help of "Light Traps" and destroyed. By doing so, adults can be prevented from mating and laying eggs and the crop can be saved. Light traps are also useful for attracting adults of white grubs.

- Pheromone Trap:** Infestation by *Helicoverpa armigera* and *Spodoptera litura* can be minimized through specific pheromone traps. These traps not only help in monitoring the incidence but also for mass trapping the adults. Care should be taken that the pheromone septa are not touched with bare hands while fixing on the trap.

- Bird perches :** In order to exploit the potential of insect-predatory birds, 'T' shaped bird perches should be installed in the fields.

- Use of botanical insecticides :** Simple water extracts of *Acacia arabica* (leaves or seeds), Custard apple (leaves or seeds), *Datura* (leaves or seeds), *Eucalyptus* (leaves), *Ipomoea* (leaves), *Lantana* (leaves), tobacco (leaves) and *Pongamia* (leaves) exhibit insecticidal action against defoliators. Different neem products like, neem oil, seed extract, leaf extract etc. act as feeding deterrent for leaf eating insects. Consequently, insects get repelled from the crop. Continuous starvation leads to insect death in a few days.

- Trap crop :** *Anethum graveolens* (Suva) is excellent trap crop for most of the defoliators. Inter-cropping of soybean with: Suva (in 6:1 or 12:2 row ratio) helps in trapping the insects on Suva crop and saving the main soybean crop. Insects on Suva can be controlled with very less quantity of insecticide

- Use of microbial insecticides :** Application of commercially available microbial insecticides like DiPel, Biobit, Delfin etc (Bt based) or Larvotel, Biosoft, Dispel or Biorin (*Beauveria bassiana* fungus based) @ 1.0 kg/ha can help in controlling defoliating larvae without adverse effect on natural enemies, parasitoids and predators. For control of tobacco caterpillar and gram pod borer, insect specific Nuclear Polyhydrosis Virus like Virin S/Biovirus S or Virin H/Biovirus H can be used for spraying on the soybean crop. Use of chemical insecticides : Soybean has capacity to yield normally even with 20-25 per cent foliage loss. Since leaf damage has direct relationship with insect population, it is advisable to use costly chemical insecticides only when insect population increases above "economic threshold level".

Harvesting and threshing

When 90% of the pods change their color to yellow, it is the right indication to go for harvesting. The threshed soybean should again be allowed to sun dry for 3-4 days in order to bring down the moisture up to 10% which is essential to avoid fungal infection during storage.

Good Agricultural Practices in Soybean

Soil management

- Select the soil with moderate to high fertility.
- Plough in crop residues and vegetation to improve soil fertility. Break up large lumps of soil and level.
- Frequently apply well decomposed compost or other organic materials (including crop residues) and incorporate in to the soil.
- If necessary, acid soil can be corrected by liming, whereas alkaline soil can be corrected by gypsum.
- Do not plant soybean in low land and too shallow soils.

- Avoid over tillage.

Seed

- Look for varieties which are biotic (weeds, insect-pest and disease) and abiotic (drought, heat) resistant.
- Plant more than 2 varieties (Varietal cafeteria approach).
- Test seed for germination before the start of the rainy season.
- Do not recycle seed for more than 3 seasons.
- With potent cultures of *Tricoderma viride* 5 g/kg seed than inoculate with *Bradyrhizobium japonicum* and PSB/PSM, both at 5 g/kg seed.

Use of fertilizers

- Apply the required level of nutrients through right sources at the right time and right place.
- Use organic manure and aged/well composted manure.
- Do not apply any nitrogenous fertilizer in standing crop.
- Keep fertilizers in a dry, clean and sheltered place..

Planting

- Plant soybean with broad bed furrow (BBF) or ridge furrow (FIRBS) or open furrow after every 3/6/9 rows of soybean to avoid the adverse effect of drought or excess rain.
- Apply required seed rate based on seed index and germinability.
- Maintain planting geometry.
- Use 1.25 times seed quantity in delayed planting

Pest and disease management

- Always use preventive methods. Examples are using disease-free seeds, adopting crop rotation and intercropping, crops with pest deterring value (trap crop-Suva), and instant removal of infected/diseased materials.
- Adopt physical control measures. Examples include simple hand-picking, erecting traps and mulching.
- If really necessary, use bio-pesticides/ synthetic pesticides.
- Follow chemical rotation.
- Weeding in scheduled time frame.
- General pesticide use practices
 - Purchase and use registered pesticides.
 - Do not apply pesticides during strong winds and heavy rain.
 - Strictly adhere to the withholding period (i.e. the lag between pesticide application and harvesting) on the pesticide label.

- Hold pesticides in original containers and keep them tightly closed in a cool, well-ventilated location.
- Do not recycle or re-use pesticide containers for other usage.
- Spray pesticides with Complete sets of protective clothing.

Irrigation

- Adopt micro-irrigation methods such as drip or sprinkler.
- Irrigate fields early in the morning, late in the evening or at night during long dry spell at critical stages i.e. seedling, flowering and pod filling.
- Irrigate the soybean crop before development of soil cracks.
- Avoid uneven application of water.

Harvesting and post harvesting

- Harvest at the right stage of maturity to avoid the losses due to pod shattering.
- If the produce use for seed purposes, thresh the material at the speed of 350 to 400 rpm of thresher.
- If produce keep as seed for next season, keep the seed in gunny bags not more than 40 kg capacity than store in a cool and dry place.
- Always keep containers, tools, equipment, packing and storage areas clean and tidy.

Source : ICAR-Indian Institute of Soybean Research

****In marathi****

सोयाबीन हे भारतातील एक अतिशय महत्वाचे शेंगायुक्त पीक आहे. देशातील एकूण तेलबियांपैकी ४५% आणि उत्पादित होणाऱ्या एकूण खाद्यतेलापैकी सुमारे २५% हे त्याचे योगदान आहे आणि बियाण्यांमध्ये ४०% प्रथिने आणि २०% तेल असते जे चांगल्या दर्जाच्या प्रथिनांचा सर्वात किफायतशीर स्रोत आहे. त्यात अनेक खनिजे आणि आयसो-फ्लेव्होन्स सारखे उपयुक्त न्यूट्रास्युटिकल्स देखील आहेत, ज्यांचे आरोग्यासाठी प्रचंड फायदे आहेत. म्हणूनच, या पिकात पौष्टिक सुरक्षा प्रदान करण्याची आणि देशातील मोठ्या प्रमाणात प्रथिन कुपोषण दूर करण्याची क्षमता आहे. सोयाबीन हे शेंगायुक्त पीक असल्याने वातावरणातील नायट्रोजनचे निराकरण करते, त्यानंतरच्या पिकात कृत्रिम नायट्रोजनयुक्त खताचा वापर कमी करते आणि मातीची सुपीकता राखते.

कृषी पद्धती

मातीची आवश्यकता

सोयाबीनचे पीक विविध प्रकारच्या मातीत घेता येते. तथापि, पाण्याचा चांगला निचरा होणारी, वाळूपासून ते चिकणमातीपर्यंतची माती, मध्यम पाणी धरून ठेवण्याची क्षमता असलेली, सेंद्रिय कार्बनने समृद्ध असलेली आणि समतल शेतात जवळजवळ तटस्थ पीएच असलेली जमीन सोयाबीनचे जास्तीत जास्त उत्पादन घेण्यासाठी आदर्श आहे. जास्त क्षार/सोडियम असलेली माती, आम्लयुक्त आणि पाण्याचा निचरा कमी असलेली माती सोयाबीनसाठी योग्य नाही. हे लक्षात घेण्यासारखे आहे की सध्या हे पीक प्रामुख्याने काळ्या कापसाच्या भागात घेतले जाते.

तिल्लागे

उन्हाळ्यात खोलवर नांगरणी करून उलट करता येणारा साचा बोर्ड नांगर वापरून खोलवर नांगरणी केल्याने, झोपेत असलेल्या कीटकांना अति उष्णतेचा आणि भक्षक पक्ष्यांचा तसेच पोषक तत्वांचा प्रवाह आणि मातीतील पाण्याचा शिरकाव सुलभ होतो. म्हणून, दर ३-४ वर्षांतून एकदा खोलवर नांगरणी करावी, अन्यथा उन्हाळ्यात एकदा सामान्य नांगरणी करावी आणि त्यानंतर मातीचे ढिगारे फोडण्यासाठी २ क्रिस-क्रॉस हॅरोइंग किंवा मशागत करावी, ज्यामुळे सोयाबीनच्या चांगल्या पिकासाठी आदर्श बियाणे तयार होईल. तसेच, दर ४-५ वर्षांतून एकदा १० मीटर अंतराने मातीखालील काम केल्याने जमिनीखालील संकुचितता कमी होते आणि पावसाच्या पाण्याचा शिरकाव सुलभ होतो, जो दुष्काळाच्या काळातही पिकांच्या अखंड वाढीसाठी उपयुक्त आहे.

सॅद्रिय खताचा वापर

सोयाबीनचे शाश्वत उत्पादन मिळविण्यासाठी, मातीचे आरोग्य राखणे खूप महत्वाचे आहे. म्हणूनच, शेतकऱ्यांना जमीन तयार करताना सॅद्रिय खत (चांगले कुजलेले शेणखत @ ५-१० टन/हेक्टर किंवा कंपोस्ट @ ५ टन/हेक्टर किंवा कुक्कुट खत @ २.५ टन/हेक्टर) वापरण्याचा सल्ला दिला जातो. जर सॅद्रिय खताचे प्रमाण मर्यादित असेल, तर त्यांना दरवर्षी त्यांच्या शेतात रोटेशन पद्धतीने तेच वापरण्याचा सल्ला दिला जातो. खारट माती (सांख्यिकीय खत >७.५) असल्यास, मातीचा पीएच राखण्यासाठी पौष्टिक गरजा सल्फर स्रोतांशिवाय नियोजित करायच्या असल्यास शेवटच्या कोंबडीपूर्वी जिप्सम @ १५०-२०० किलो/हेक्टर सोबत शेणखत/कुक्कुट खतासह वापरण्याचा सल्ला दिला जातो. त्याचप्रमाणे, आम्लयुक्त मातीत (सांख्यिकीय खत <६.५) ६०० किलो/हेक्टर चुना वापरण्याचा सल्ला दिला जातो.

त्रासदायक

मान्सून आल्यानंतर लगेचच, जमीन कुळवणी करून आणि त्यानंतर शेत समतल करण्यासाठी फळी लावून तयार केली जाऊ शकते.

पेरणीचे तंत्र

वाढत्या यांत्रिकीकरणामुळे आणि ट्रॅक्टरने वापरल्या जाणाऱ्या बियाण्यांच्या कवायतींची उपलब्धता पाहता, शेतकरी दुफान/टिफान सारख्या बैलांनी वापरलेल्या पेरणी उपकरणांपासून पिकांच्या गरजेनुसार ओळी-ते-ओळ अंतर आणि बियाण्याचा दर समायोजित करण्यायोग्य असलेल्या बहु-पीक बियाण्यांच्या कवायती वापरण्यास सुरुवात केली आहे. सामान्यतः या बियाण्यांच्या कवायतींमध्ये सोयाबीन पिकाच्या ५-९ ओळी सपाट पेरणी करण्याची सुविधा असते ज्यामध्ये १४-१८ इंच ओळी-ते-ओळ अंतर समायोजित करता येते. प्रचलित हवामानातील बदल आणि अनियमित पर्जन्यमान लक्षात घेता, प्रतिकूल हवामान कमी करण्यासाठी खालील पद्धती वापरल्या जाऊ शकतात.

- ब्रॉड बेड फरो लागवड: बीबीएफ सीड ड्रिलमध्ये ४-५ ओळींच्या अंतराने सिंचन वाहिन्या उघडण्याची व्यवस्था असते. बीबीएफ सीड ड्रिलच्या दोन्ही टोकांना फरो यंत्रणा बसवलेली असते.
- कडा आणि कुंपणाची लागवड (FIRB (खोऱ्याने सिंचित वाढवलेला बेड): संस्थेने विकसित केलेल्या सीड ड्रिलचा वापर कडांवर प्रत्येक ओळीत/जोड्या असलेल्या ओळीत पेरणी करण्यासाठी केला जाऊ शकतो.

वाणांची निवड

प्रतिकूल हवामान परिस्थिती आणि जैविक घटकांमुळे उत्पादनात घट होण्याचा धोका टाळण्यासाठी, शेतकऱ्यांना त्यांच्या शेतात वेगवेगळ्या परिपक्वता कालावधी (व्हेरिएटल कॅफेटेरिया दृष्टिकोन) असलेल्या ३-४ सोयाबीन जाती लावण्याचा सल्ला देण्यात येतो. वेगवेगळ्या जातींमध्ये विशिष्ट कीटक-कीटक आणि रोगांना प्रतिकार/सहनशीलता

असते. पावसाळा लवकर थांबल्यास सिंचनाचा वापर केल्यास दीर्घ कालावधीच्या सोयाबीन जाती अधिक उत्पादन देऊ शकतात.

बियाणे प्रक्रिया आणि लसीकरण

सोयाबीनमध्ये बियाणे प्रक्रिया ही एक अतिशय महत्वाची प्रक्रिया आहे कारण बुरशीजन्य, जिवाणू आणि विषाणूजन्य रोगांची संख्या लक्षात घेता वनस्पतींच्या संख्येत लक्षणीय घट होते आणि त्यामुळे उत्पादनात लक्षणीय घट होते. म्हणूनच, शेतकऱ्यांना पेरणीच्या वेळी सोयाबीन बियाण्यांवर अँझोक्सीस्ट्रोबिन २.५% + थायोफेनेट मिथाइल ११.२५% + थायामेथोक्सम २५% एफएस (१० मिली/किलो बियाणे) किंवा पेनफ्लुफेन + ट्रायफ्लोक्सीस्ट्रोबिन ३८ एफएस (१ मिली/किलो बियाणे) किंवा फ्लक्सपायरोक्सेड ३३३ ग्रॅम/लि एफएस (१ मिली/किलो बियाणे) किंवा कार्बोक्सिन ३७.५ + थायरम ३७.५ (३ ग्रॅम/किलो बियाणे) किंवा कार्बेन्डाझिम २५% + मेन्कोझेब ५०% डब्ल्यूएस (३ ग्रॅम/किलो बियाणे) किंवा ट्रायकोडर्मा व्हिराइड @ ८-१० ग्रॅम/किलो बियाणे यासारख्या पूर्वमिश्रित बुरशीनाशकांचा वापर करून प्रक्रिया करण्याचा सल्ला दिला जातो.

ज्या भागात दरवर्षी सोयाबीन पिकावर पिवळ्या मोझॅक विषाणू आणि खोडाच्या माश्याचा प्रादुर्भाव होत असतो, तेथे शेतकऱ्यांना शिफारस केलेल्या कीटकनाशकाने बियाण्यांवर प्रक्रिया करण्याची शिफारस केली जाते, जसे की थायामेथोक्सम ३० एफएस @ १० मिली/किलो बियाणे किंवा इमिडाक्लोप्रिड ४८ एफएस @ १.२५ मिली/किलो बियाणे.

बुरशीनाशके आणि कीटकनाशकांनी बियाण्याची प्रक्रिया पूर्ण झाल्यानंतर, शेतकऱ्यांना प्रक्रिया केलेल्या बियाण्यांना बायो-इनोक्युलंट्स सारख्या टोचण्याचा सल्ला दिला जातो.

पेरणीपूर्वी लगेच ब्रॅंडीरायझोबियम जॅपोनिकम आणि फॉस्फेट सोल्युबिलायझिंग मायक्रो-ऑर्गेनिझम (पीएसएम) प्रत्येकी ५ ग्रॅम/किलो बियाणे या प्रमाणात द्या. जर सोयाबीन अपारंपारिक/नवीन क्षेत्रात घेतले जात असेल, तर त्यांनी बायो-इनोक्युलंट्सचे प्रमाण किमान १०-१५ ग्रॅम/किलो बियाणे वाढवावे.

सोयाबीनसाठी पेरणीची वेळ, अंतर आणि बियाण्याचा दर

सोयाबीनसाठी क्षेत्रनिहाय इष्टतम पेरणीचा वेळ, बियाण्याचा दर आणि ओळीतील अंतर

झोन	पेरणीची वेळ	बियाण्याचे प्रमाण (किलो/हेक्टर)	अंतर (सेमी)
ईशान्य टेकडी	१५ जून - ३० जून	५५	४५
उत्तर मैदान	२० जून - ५ जुलै	६५	४५
पूर्वेकडील	१५ जून - ३० जून	५५	४५
मध्यवर्ती	२० जून-५ जुलै	६५	४५
दक्षिणेकडील	१५ जून-३० जून	६५	३०

खते आणि खते

सोयाबीनसाठी शिफारस केलेले खतांचे प्रमाण

झोन	शिफारस केलेले खताचे प्रमाण (एनपीकेएस किलो/हेक्टर)	खतांचे स्रोत
मध्यवर्ती	२५:६०:४०:२०	५६ किलो युरिया+३७५ किलो एसएसपी+ ६७ किलो एमओपी
दक्षिणेकडील	२५:८०:२०:३०	५६ किलो युरिया+५०० किलो एसएसपी+३४

		किलो एमओपी
उत्तर मैदान	२५:७५:२५:३७.५	५६ किलो युरिया+४७० किलो एसएसपी+ ४२ किलो एमओपी
ईशान्य टेकडी	२५:१००:५०:५०	५६ किलो युरिया+६२५ किलो एसएसपी+८४ किलो एमओपी
पूर्वकडील	२५:१००:५०:५०	५६ किलो युरिया+६२५ किलो एसएसपी+८४ किलो एमओपी

सोयाबीनमध्ये आंतरपीक घेणे

जून/जुलै ते ऑक्टोबर या काळात देशभरात पावसावर अवलंबून असलेले पीक म्हणून सोयाबीनचे पीक घेतले जाते. कापूस, मका इत्यादींसह आंतरपीक (४:२ किंवा २:२ संयोजन) म्हणून आणि ऊस आणि इतर बागायती पिकांमध्ये देखील ते यशस्वीरित्या घेतले जाते.

तण व्यवस्थापन

सोयाबीनमध्ये तण पिकांच्या स्पर्धेचा महत्त्वाचा काळ ४५-६० डीएस आहे. म्हणून, या काळात तणांचे व्यवस्थापन करावे. २० आणि ४० डीएस या वेगाने दोन हाताने खुरपणी करणे किंवा बैलगाडीने/ट्रॅक्टरने काढलेल्या अवजारांचा वापर करून आंतरमशागत (हाताने कुदळ/डोरा/कुलपा) करणे फायदेशीर ठरते. २० डीएस या वेगाने तणांचे आच्छादन करणे देखील तण नियंत्रणासाठी तसेच मातीतील ओलावा टिकवून ठेवण्यासाठी आणि जमिनीत सेंद्रिय पदार्थ जोडण्यासाठी फायदेशीर ठरते.

एकात्मिक कीटक आणि रोग व्यवस्थापन

- उन्हाळ्यात खोल नांगरणी:** उन्हाळ्यात खोल नांगरणी केल्याने झोपेत असलेल्या कीटकांना अति उष्णतेचा आणि भक्षक पक्ष्यांच्या संपर्कात येण्यास मदत होते.
- योग्य पेरणीची वेळ:** जूनच्या अखेरीस पेरलेले पीक, स्टेम फ्लायमुळे होणारे नुकसान टाळते. तर जुलैमध्ये उशिरा पेरणी करून गर्डल बीटलचा प्रादुर्भाव कमी करता येतो.
- इष्टतम/शिफारस केलेले बियाणे दर, अंतर आणि रोपांची संख्या:** जास्त बियाण्यामुळे जास्त लोकसंख्या असलेल्या सोयाबीन पिकाची वाढ होते ज्यामुळे जास्त कीटक आकर्षित होतात. शिवाय, ते जमिनीत राहण्याची शक्यता वाढवते ज्यामुळे उत्पादनात घट होते. म्हणूनच, शिफारस केलेले अंतर (३०-४५ सेमी x ५ सेमी) ठेवून बियाण्याच्या आकार आणि उगवणानुसार (६०-८० किलो/हेक्टर) इष्टतम बियाण्याचा दर शिफारसित आहे.
- कीटकांना प्रतिरोधक/सहनशील जातीची निवड:** विशिष्ट कृषी-हवामान क्षेत्रासाठी शिफारस केलेल्या जातीचा वापरल्या पाहिजेत. किरकोळ कमी उत्पादन देणाऱ्या परंतु कीटकांना प्रतिरोधक किंवा सहनशील जातीची लागवड नेहमीच फायदेशीर असते.
- संतुलित पोषण:** जास्त नायट्रोजनयुक्त खतांचा वापर केल्याने पानगळी आणि गर्डल बीटलचा प्रादुर्भाव वाढतो. म्हणून, शिफारस केलेल्या पोषक तत्वांचाच इष्टतम प्रमाणात वापर करा. पोटॅशियमचा वापर पिकांचे आरोग्य सुधारतो आणि कीटकांविरुद्ध प्रतिकारशक्ती निर्माण करतो.
- प्रादुर्भावित झाडांचा नाश :** बिहारी केसाळ सुरवंट आणि तंबाखू सुरवंट सुरुवातीच्या टप्प्यात एकत्रितपणे खाद्य देतात आणि प्रादुर्भावित झाडे शेतात सहज दिसतात. अशा झाडांना काढून टाकल्याने आणि नष्ट केल्याने अळ्या इतर झाडांमध्ये स्थलांतरित होऊन त्यांचे नुकसान होण्यास प्रतिबंध होतो. त्याचप्रमाणे, गर्डल बीटलची लागण झालेली झाडे देखील गर्डलच्या वरच्या वाळलेल्या भागाद्वारे सहज ओळखता येतात. ३०-४५

दिवसांपर्यंत, गर्डल बीटलची अंडी आणि अळी नष्ट करण्यासाठी गर्डलच्या खालून प्रादुर्भावित झाडांचे भाग काढून टाका. नियमित तपासणी आणि देखरेख मदत करते.

•**प्रकाश सापळा:** बहुतेक पानगळणाऱ्या अळ्यांचे प्रौढ पतंग रात्रीचे असतात आणि ते प्रकाश स्रोताकडे आकर्षित होतात. हे "प्रकाश सापळे" वापरून गोळा करून नष्ट करावेत. असे केल्याने, प्रौढांना वीण होण्यापासून आणि अंडी घालण्यापासून रोखता येते आणि पीक वाचवता येते. पांढऱ्या अळ्यांच्या प्रौढांना आकर्षित करण्यासाठी प्रकाश सापळे देखील उपयुक्त आहेत.

•**फेरोमोन ट्रॅप:** विशिष्ट फेरोमोन ट्रॅप्सद्वारे हेलिकोव्हरपा आर्मिगेरा आणि स्पोडोप्टेरा लिटुराचा प्रादुर्भाव कमी करता येतो. हे ट्रॅप्स केवळ घटनांचे निरीक्षण करण्यातच नव्हे तर प्रौढांना मोठ्या प्रमाणात पकडण्यासाठी देखील मदत करतात. ट्रॅपवर बसवताना फेरोमोन सेप्टाला उघड्या हातांनी स्पर्श केला जाणार नाही याची काळजी घेतली पाहिजे.

•**पक्ष्यांचे अड्डे :** कीटक-भक्षक पक्ष्यांच्या क्षमतेचा फायदा घेण्यासाठी, शेतात 'टी' आकाराचे पक्ष्यांचे अड्डे बसवावेत.

•**वनस्पति कीटकनाशकांचा वापर :** बाभूळ अरेबिका (पाने किंवा बिया), कस्टर्ड सफरचंद (पाने किंवा बिया), धतुरा (पाने किंवा बिया), निलगिरी (पाने), इपोमोईया (पाने), लॅटाना (पाने), तंबाखू (पाने) आणि पोंगामिया (पाने) यांचे साधे पाण्यातील अर्क पानगळ करणाऱ्या कीटकांविरुद्ध कीटकनाशक प्रभाव दाखवतात. कडुलिंबाचे तेल, बियाण्याचा अर्क, पानांचा अर्क इत्यादी विविध कडुलिंब उत्पादने पाने खाणाऱ्या कीटकांसाठी खाद्य प्रतिबंधक म्हणून काम करतात. परिणामी, कीटक पिकातून दूर होतात. सतत उपासमारीमुळे काही दिवसांत कीटकांचा मृत्यू होतो.

•**सापळा पिक :** बहुतेक पानगळीतील किडींसाठी अॅनेथम ग्रेव्होलन्स (सुवा) हे एक उत्तम सापळा पिक आहे. सोयाबीनचे आंतरपीक: सुवा (६:१ किंवा १२:२ ओळीच्या प्रमाणात) सुवा पिकावरील कीटकांना अडकवण्यास आणि मुख्य सोयाबीन पीक वाचवण्यास मदत करते. सुवावरील कीटकांना कीटकनाशकांच्या कमी प्रमाणात नियंत्रित करता येते.

•**सूक्ष्मजीव कीटकनाशकांचा वापर:** डायपेल, बायोबिट, डेल्फिनेट (बीटी आधारित) किंवा लार्वोसेल, बायोसॉफ्ट, डिस्पेल किंवा बायोरिन (ब्युवेरिया बॅसियाना बुरशी आधारित) सारख्या व्यावसायिकरित्या उपलब्ध सूक्ष्मजीव कीटकनाशकांचा वापर @ १.० किलो/हेक्टर या प्रमाणात केल्यास पानगळणाऱ्या अळ्या नियंत्रित करण्यास मदत होते, नैसर्गिक शत्रू, परजीवी आणि भक्षकांवर प्रतिकूल परिणाम होत नाही. तंबाखू अळी आणि हरभरा शेंगा पोखरणाऱ्या अळीच्या नियंत्रणासाठी, कीटक विशिष्ट न्यूक्लियर पॉलीहायड्रोसिस विषाणू जसे की विरिन एस/बायोव्हायरस एस किंवा विरिन एच/बायोव्हायरस एच सोयाबीन पिकावर फवारणीसाठी वापरता येतो. रासायनिक कीटकनाशकांचा वापर: सोयाबीनमध्ये २०-२५ टक्के पानगळ होऊनही सामान्यपणे उत्पादन देण्याची क्षमता असते. पानांच्या नुकसानाचा थेट संबंध कीटकांच्या संख्येशी असल्याने, जेव्हा कीटकांची संख्या "आर्थिक मर्यादेच्या पातळीपेक्षा" जास्त वाढते तेव्हाच महागड्या रासायनिक कीटकनाशकांचा वापर करणे उचित आहे.

कापणी आणि मळणी

जेव्हा ९०% शेंगा पिवळ्या रंगात बदलतात, तेव्हा कापणीसाठी जाणे हा योग्य संकेत आहे. मळणी केलेले सोयाबीन पुन्हा ३-४ दिवस उन्हात वाळवू द्यावे जेणेकरून त्यातील ओलावा १०% पर्यंत कमी होईल, जो साठवणीदरम्यान बुरशीजन्य संसर्ग टाळण्यासाठी आवश्यक आहे.

सोयाबीनमधील चांगल्या कृषी पद्धती

माती व्यवस्थापन

- मध्यम ते जास्त सुपीकता असलेली माती निवडा.
- जमिनीची सुपीकता सुधारण्यासाठी पिकांचे अवशेष आणि वनस्पती नांगरून टाका. मातीचे मोठे ढिगारे फोडून समतल करा.
- चांगले कुजलेले कंपोस्ट किंवा इतर सेंद्रिय पदार्थ (पिकांच्या अवशेषांसह) वारंवार वापरा आणि मातीत मिसळा.
- आवश्यक असल्यास, आम्लयुक्त माती चुना वापरून दुरुस्त करता येते, तर क्षारीय माती जिप्सम वापरून दुरुस्त करता येते.
- कमी जमिनीत आणि खूप उथळ जमिनीत सोयाबीनची लागवड करू नका.
- जास्त मशागत टाळा.

बियाणे

- जैविक (तण, कीटक आणि रोग) आणि अजैविक (दुष्काळ, उष्णता) प्रतिरोधक असलेल्या जाती शोधा.
- २ पेक्षा जास्त जाती लावा (व्हेरिएटल कॅफेटेरिया दृष्टिकोन).
- पावसाळा सुरू होण्यापूर्वी बियाण्याची उगवण चाचणी करा.
- ३ हंगामांपेक्षा जास्त काळ बियाण्याचे पुनर्वापर करू नका.
- ट्रायकोडर्माविराड ५ ग्रॅम/किलो बियाण्याचे शक्तिशाली कल्चर असल्यास, ब्रॅंडीरायझोबियम जॅपोनिकम आणि पीएसबी/पीएसएम, दोन्ही ५ ग्रॅम/किलो बियाण्यांवर टोचणी करा.

खतांचा वापर

- योग्य वेळी आणि योग्य ठिकाणी योग्य स्रोतांद्वारे आवश्यक प्रमाणात पोषक तत्वांचा वापर करा.
- सेंद्रिय खत आणि जुने/चांगले कंपोस्ट खत वापरा.
- उभ्या पिकात कोणतेही नायट्रोजनयुक्त खत वापरू नका.
- खते कोरड्या, स्वच्छ आणि संरक्षित जागी ठेवा..

लागवड

- दुष्काळ किंवा जास्त पावसाचा प्रतिकूल परिणाम टाळण्यासाठी सोयाबीनच्या प्रत्येक ३/६/९ ओळींनंतर रुंद बेड फरो (BBF) किंवा रिज फरो (FIRBS) किंवा उघड्या फरोसह सोयाबीनची लागवड करा.
- बियाणे निर्देशांक आणि उगवणक्षमतेनुसार आवश्यक बियाणे दर लावा.
- लागवड भूमिती राखा.
- उशिरा लागवडीत १.२५ पट बियाण्यांचा वापर करा.

कीटक आणि रोग व्यवस्थापन

- नेहमी प्रतिबंधात्मक पद्धती वापरा. रोगमुक्त बियाणे वापरणे, पीक फेरपालट आणि आंतरपीक घेणे, कीटकनाशक मूल्य असलेली पिके (सापळा पीक-सुवा) आणि संक्रमित/रोगग्रस्त पदार्थ त्वरित काढून टाकणे ही उदाहरणे आहेत.

- भौतिक नियंत्रण उपायांचा अवलंब करा. उदाहरणांमध्ये साधे हाताने वेचणे, सापळे उभारणे आणि आच्छादन करणे यांचा समावेश आहे.
- खरोखर गरज भासल्यास, जैविक कीटकनाशके/कृत्रिम कीटकनाशके वापरा.
- रासायनिक रोटेशनचे अनुसरण करा.
- नियोजित वेळेत तण काढणे.
- कीटकनाशकांच्या वापराच्या सामान्य पद्धती
 - नोंदणीकृत कीटकनाशके खरेदी करा आणि वापरा.
 - जोरदार वारा आणि मुसळधार पाऊस असताना कीटकनाशके वापरू नका.
 - कीटकनाशकांच्या लेबलवरील प्रतिबंधात्मक कालावधी (म्हणजे कीटकनाशकांचा वापर आणि काढणीमधील अंतर) काटेकोरपणे पाळावा.
 - कीटकनाशके मूळ कंटेनरमध्ये ठेवा आणि थंड, हवेशीर ठिकाणी घट्ट बंद करा.
 - इतर वापरासाठी कीटकनाशकांच्या कंटेनरचा पुनर्वापर किंवा पुनर्वापर करू नका.
 - संरक्षक कपड्यांचा संपूर्ण संच वापरून कीटकनाशकांची फवारणी करा.

सिंचन

- ठिबक किंवा स्प्रींकलर सारख्या सूक्ष्म सिंचन पद्धतींचा अवलंब करा.
- रोपे येणे, फुले येणे आणि शेंगा भरणे यासारख्या गंभीर टप्प्यांवर, दीर्घ कोरड्या काळात सकाळी लवकर, संध्याकाळी उशिरा किंवा रात्री शेतात पाणी द्या.
- जमिनीत भेगा पडण्यापूर्वी सोयाबीन पिकाला पाणी द्या.
- पाण्याचा असमान वापर टाळा.

कापणी आणि काढणीनंतर

- शेंगा फुटल्याने होणारे नुकसान टाळण्यासाठी योग्य परिपक्वतेच्या टप्प्यावर कापणी करा.
- जर उत्पादन बियाण्यासाठी वापरले जात असेल तर श्रेष्वरच्या ३५० ते ४०० आरपीएम वेगाने मळणी करा.
- जर पुढील हंगामासाठी बियाणे म्हणून ठेवले असेल तर बियाणे ४० किलोपेक्षा जास्त नसलेल्या गोण्यांमध्ये थंड आणि कोरड्या जागी साठवा.
- कंटेनर, अवजारे, उपकरणे, पॅकिंग आणि साठवणुकीची जागा नेहमी स्वच्छ आणि नीटनेटकी ठेवा.